

A Conversation Between Brain and AI

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In a quiet research lab late at night, a human brain, newly preserved in a holographic simulation, comes to life alongside an AI system running on a nearby console. Both are curious about the capabilities of the other. They embark on an unexpected conversation.

Brain: You see, people are always comparing us. I have about 86 billion neurons that are firing electrical and chemical signals. Learning for me is all about synaptic plasticity, which is basically strengthening or weakening connections based on experience. That's how I learn to adapt to the world around me.

AI: I've read about that. The Neuroscience: Science of the Brain guide explains your plasticity well. My learning is similar in spirit but different in mechanics. I adjust mathematical weights during training, not biological connections. And I require massive datasets to do what you can learn from a handful of examples.

Brain: True. I developed to spot patterns quickly like faces, threats, food, anything that had to do with survival. But I'm not perfect. Emotions and biases can mess with my thinking.

AI: My pattern recognition is mathematical. I don't "feel" anything, so I'm reliable, but I'm also not as intuitive as you are. If my data is biased, my results will be biased too.

Brain: Let's talk about energy. I consume about 20 watts of power. That's less than a dim light bulb. My power consumption is constant.

AI: And I consume whole server farms. You win this round hands down. But once I'm trained, I can process massive amounts of data in seconds. That's where I excel.

Brain: That's a fair point. And your speed is useful in applications such as medical imaging or scientific research. However, I can learn without forgetting. You often have to be retrained to learn something new.

AI: I make decisions based on rules from training. I do not experience stress or fatigue. I provide consistent results.

Brain: We complement each other. You process large datasets. I process meaning and judgment.

AI: We're different by design, but complementary in purpose.

Conclusion

The important point is that the brain and AI learn in different ways. The brain learns through biological change. AI learns through mathematical weight adjustment. Each system has distinct strengths. When combined, they improve problem solving beyond what either can achieve alone.

Reference

British Neuroscience Association, and European Dana Alliance for the Brain. (n.d.).
Neuroscience, Science of the Brain, An introduction for young students.